

Reproducing DistilBERT: Verifying Efficiency-Performance Tradeoffs on IMDB Sentiment Classification

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1. Introduction

Large pretrained language models such as BERT achieve strong performance across many natural language processing tasks, but they are expensive to train, store, and deploy. BERT-base has about 110 million parameters, which makes it slower and more computationally demanding, especially when latency, memory, or cost matters.

Our project reproduces key results from DistilBERT, a compressed version of BERT introduced by Sanh, Debut, Chaumond, and Wolf. DistilBERT uses knowledge distillation, where a smaller student model learns from a larger teacher model. It reduces BERT-base from 12 Transformer layers to 6, removes token-type embeddings and the pooler, and keeps most of BERT’s language understanding ability while being smaller and faster.

The original paper claims that DistilBERT retains about 97% of BERT’s performance while being 40% smaller and 60% faster. Our goal was to test this claim on IMDB sentiment classification and compare DistilBERT against BERT-base using accuracy, model size, inference speed, and training time.

2. Chosen Result

We reproduced the IMDB binary sentiment classification result from the DistilBERT paper, along with its efficiency comparison. The paper reports that DistilBERT achieves 92.82% test accuracy on IMDB, while BERT-base achieves 93.46% – a 0.64 percentage point difference.

We chose IMDB because it directly tests the purpose of the paper – if a smaller distilled model can preserve most of the performance of a larger model. IMDB is also practical because it is a standard sentiment classification dataset and does not require reproducing the full GLUE benchmark or pretraining DistilBERT from scratch.

3. Methodology

We fine-tuned distilbert-base-uncased and bert-base-uncased from HuggingFace Transformers on the full IMDB dataset, which contains 25,000 training examples and 25,000 test examples. We did not pretrain DistilBERT from scratch because the original paper required 8 V100 GPUs for about 90 hours. Instead, we used the released pretrained DistilBERT weights.

Both models were trained under the same conditions – 3 epochs, learning rate of $2e-5$, batch size of 32, maximum sequence length of 512, weight decay of 0.01, warmup ratio of 0.1, and fp16 mixed precision on CUDA. We measured test accuracy, parameter count, FP32 model size, GPU inference latency, CPU inference latency, and training time.

We varied the maximum sequence length to 32, 64, 128, and 256 as well, because it is impossible to use a larger maximum sequence length of 512 on mobile phones.

4. Results and Analysis

Our results support the main claims of the DistilBERT paper. DistilBERT retained almost all of BERT-base’s accuracy while being significantly smaller and faster.

Metric	DistilBERT	BERT-base	Our Result	Paper Claim
IMDb Test Accuracy	93.06%	94.11%	98.9% retained	97% retained
Parameters	67.0M	109.5M	38.8% fewer	40% fewer
Model Size, FP32	255 MB	418 MB	38.8% smaller	Not reported
GPU Inference Latency	6.58 ms	12.46 ms	1.89x faster	Not reported
CPU Inference Latency	481.9 ms	953.5 ms	1.98x faster	1.63x faster
Training Time	19.5 min	38.6 min	1.98x faster	Not reported

DistilBERT achieved 93.06% accuracy compared to 94.11% for BERT-base, meaning it retained 98.9% of BERT-base’s accuracy. The accuracy gap was only 1.05 percentage points, showing that the smaller model preserved most of the larger model’s performance.

The parameter results also closely matched the paper. DistilBERT had 67.0 million parameters compared to 109.5 million for BERT-base, a 38.8% reduction. This is very close to the paper’s 40% smaller claim.

The speed results were stronger than expected. DistilBERT was 1.89x faster on GPU and 1.98x faster on CPU. This suggests that its efficiency comes mainly from its simpler architecture, especially using 6 Transformer layers instead of 12.

5. Reflection

This project showed that knowledge distillation is an effective way to compress large language models without losing much accuracy. DistilBERT was almost twice as fast as BERT-base while losing only about one percentage point of accuracy on IMDb.

The main takeaway is that smaller models can be much more practical for real-world deployment. In settings where speed, memory, and cost matter, DistilBERT provides a strong tradeoff between performance and efficiency.

Future work could test DistilBERT on additional GLUE tasks, apply task-specific distillation during fine-tuning, use quantization to reduce model size further, or compare distillation results on newer models such as RoBERTa.

6. References

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GitHub: <https://github.com/EcSky19/DistilBERT-Replication>